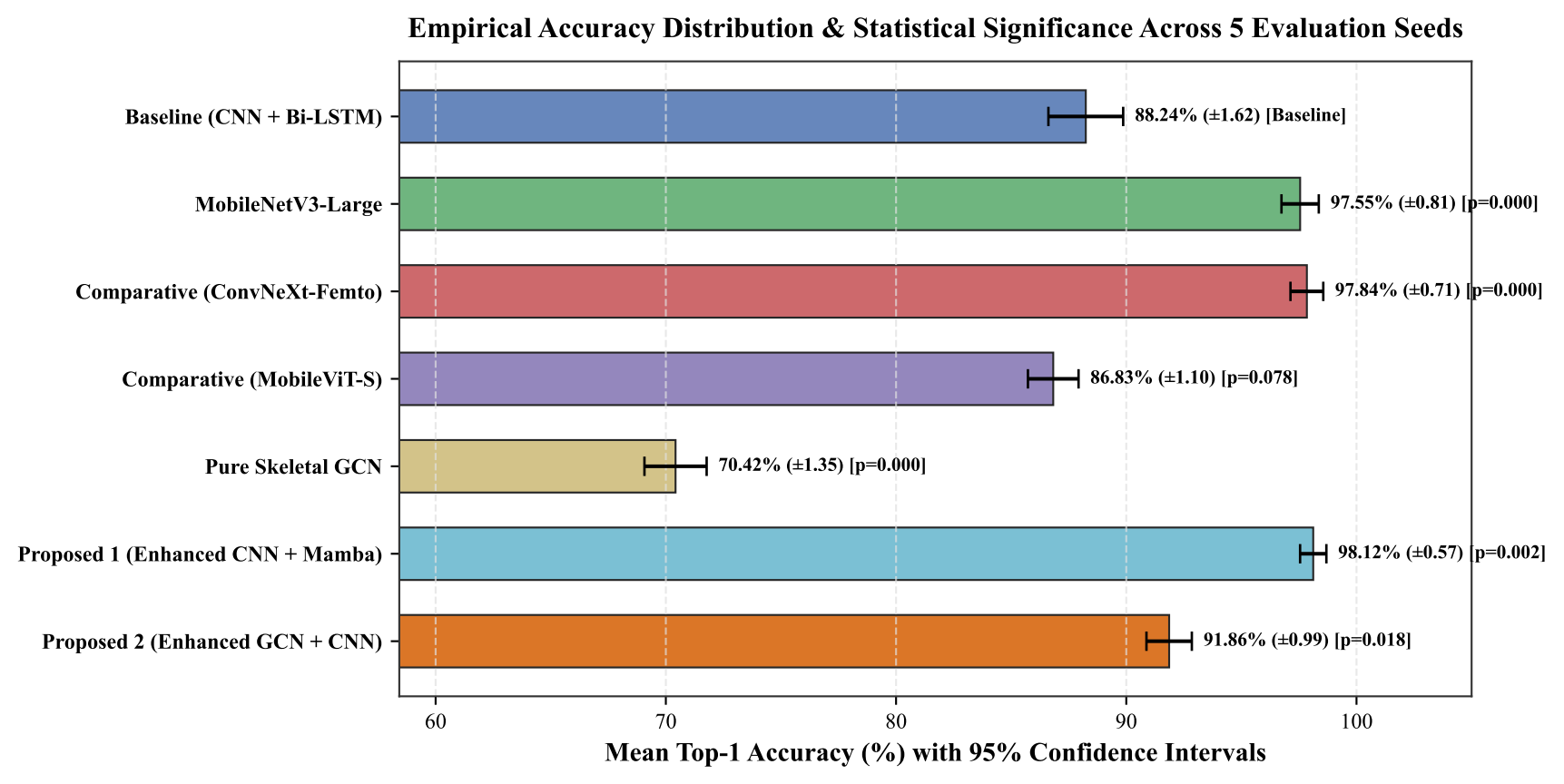
Empirical Accuracy Distribution & Statistical Significance Across 5 Evaluation Seeds
Baseline (CNN + Bi-LSTM)
88.24% (±1.62) [Baseline]
MobileNetV3-Large
97.55% (±0.81) [p=0.000]
Comparative (ConvNeXt-Femto)
97.84% (±0.71) [p=0.000]
Comparative (MobileViT-S)
86.83% (±1.10) [p=0.078]
Pure Skeletal GCN
70.42% (±1.35) [p=0.000]
Proposed 1 (Enhanced CNN + Mamba)
98.12% (±0.57) [p=0.002]
Proposed 2 (Enhanced GCN + CNN)
91.86% (±0.99) [p=0.018]
60
70
80
90
100
Mean Top-1 Accuracy (%) with 95% Confidence Intervals